Brightlearn Activity
Exercise 2: SQL Aggregate Functions & Operators

Section A: The Students Table

1. Query:

```
SELECT DISTINCT department
FROM students;
```

Output:

department
IT
HR
Finance

2. Query:

```
SELECT department,
       AVG(age) AS avg_age
FROM students
GROUP BY department;
```

Output:

department	avg_age
IT	20.5
HR	22
Finance	23

3. Query:

```
SELECT department,
       COUNT(student_id) AS student_count
FROM students
GROUP BY department
~~HAVING student_count > 1;~~ HAVING COUNT(student_id) > 1;
```

Output:

department	student_count
IT	2
HR	2

4. Query:

```
SELECT *
FROM students
WHERE age BETWEEN 21 AND 23;
```

Output:

student_id	name	age	department
2	Bob	22	HR
3	charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

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5. Query:

```
SELECT *
FROM students
WHERE department IN('IT','HR') AND age>21;
```

Output

Student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

Section B: The Courses Table

6. Query:

```
SELECT department,
       SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(credits)>5;
```

Output

department	total_credits
IT	11

7. Query:

```
SELECT *
FROM courses
WHERE NOT credits=4;
```

Output

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. Query:

```
SELECT course_id,
       course_name, credits
FROM courses
~~GROUP B~~ ORDER BY credits DESC
LIMIT 3;
```

Output

Course_id	course_name	Credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

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Section C: The enrollment Table

9. Query:

```
SELECT MAX(grade) AS max_grade,
       MIN(grade) AS min_grade,
       AVG(grade) AS avg_grade
FROM enrollments;
```

Output:

max_grade	min_grade	avg_grade
90	78	84.6

10. Query:

```
SELECT course_id,
       COUNT(enrollment_id) AS enrollment_count
FROM enrollments;
~~Group~~
GROUP BY course_id;
```

Output:

Course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

~~11. Query~~

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Section D: The Salaries Table

11. Query

```
SELECT department,
       SUM(Salary) AS total_Salary,
       SUM(bonus) AS total_bonus,
FROM salaries
GROUP BY department;
```

Output:

department	total_salary	total bonus
IT	122 000	10 500
HR	109 000	6000
Finance	70 000	7 500

12. Query:

```
SELECT department,
       AVG(Salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(Salary) > 55 000;
```

Output:

department	avg_salary
IT	61 000
Finance	70 000

13. Query:

```
SELECT employee_id,
       name,
       ~~depart~~ Salary,
       bonus,
       (Salary + bonus) AS total_compensation
FROM salaries
WHERE (salary + bonus) > 60,000;
```

Output:

employee-id	name	Salary	bonus	total Compensation
1	Tom	60 000	5000	65 000
3	Spike	70 000	6000	76 000
4	Tyke	62 000	55 000	67 500

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Section E: The projects Table

14. Query:

```
SELECT department,
       SUM(budget) AS total_budget,
       AVG(budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG(budget) > 70 000;
```

Output:

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15. Query:

```
SELECT project_id,
       project_name,
       department,
       budget
FROM projects
WHERE budget BETWEEN 50 000 AND 120 000
      AND NOT IN ('Marketing');
```

Output:

Project_id	project_name	department	budget
1	AI APP	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000

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